

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab : Embeded Systems
Semester	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	08
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	Implementation of a Real-Time Temperature Monitoring and Fan Speed Control System Using ESP32 and FreeRTOS.
Theory (100 words)	Real-time temperature monitoring and fan speed control using ESP32 and FreeRTOS is an embedded system application designed to continuously measure temperature and automatically control the speed of a cooling fan. The ESP32 is a powerful microcontroller with built-in processing capabilities and GPIO interfaces, making it suitable for real-time monitoring and control applications. A temperature sensor is used to measure the surrounding temperature and provide the corresponding sensor value to the ESP32. The ESP32 processes this value and compares it with predefined temperature limits. Based on the measured temperature, the speed of the fan can be increased or decreased. PWM (Pulse Width Modulation) is used to control the fan speed by varying the duty cycle of the PWM signal. A higher duty cycle provides greater power to the fan, resulting in higher speed, while a lower duty cycle reduces the fan speed. FreeRTOS (Free Real-Time Operating System) is used to manage the different operations of the system as independent tasks. One task can continuously read the temperature, while another task can control the fan according to the measured temperature. FreeRTOS provides task scheduling and timing mechanisms that allow these operations to execute efficiently and predictably. The tasks can communicate using shared

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	variables or synchronization mechanisms when required. The system therefore provides continuous temperature monitoring and automatic fan speed adjustment. As the temperature increases, the fan speed is increased to provide more cooling, and when the temperature decreases, the fan speed is reduced. This demonstrates the use of ESP32, sensors, PWM, and FreeRTOS for developing a simple real-time embedded control system.
Procedure and Execution (100rds)	1. Open Wokwi in a web browser and create a new ESP32 project by selecting the ESP32 DevKit board. 2. Add the required components, including an ESP32 DevKit, DHT22 temperature sensor, DC motor, NPN transistor, resistor, and diode. 3. Connect the DHT22 sensor to the ESP32 by connecting VCC to 3.3V, GND to GND, and the DATA pin to GPIO 15. 4. Connect the DC motor to the ESP32 through the NPN transistor. Connect the transistor emitter to GND, the motor to the collector, and the transistor base to an ESP32 PWM pin through a resistor. 5. Connect a diode across the motor terminals to protect the circuit from voltage spikes generated by the motor. 6. Open the Library Manager in Wokwi and add the required DHT sensor library. 7. Write the ESP32 program using FreeRTOS, creating one task for continuously reading the temperature and another task for controlling the fan speed. 8. Configure the selected ESP32 GPIO pin for PWM output so that the fan speed can be controlled by changing the PWM duty cycle. 9. Set suitable temperature ranges in the program so that the fan remains OFF or runs at low speed at lower temperatures and increases its speed as the temperature rises. 10. Compile the program in Wokwi and correct any errors displayed by the compiler. 11. Click the Start Simulation button to execute the program. 12. During execution, the ESP32 continuously reads the temperature from the DHT22 sensor using the temperature-monitoring FreeRTOS task. 13. The measured temperature is displayed on the Serial Monitor along with the corresponding fan speed or PWM value. 14. The fan-control task compares the measured temperature with the predefined temperature ranges and changes the PWM duty cycle accordingly. 15. Increase or decrease the DHT22 temperature value during

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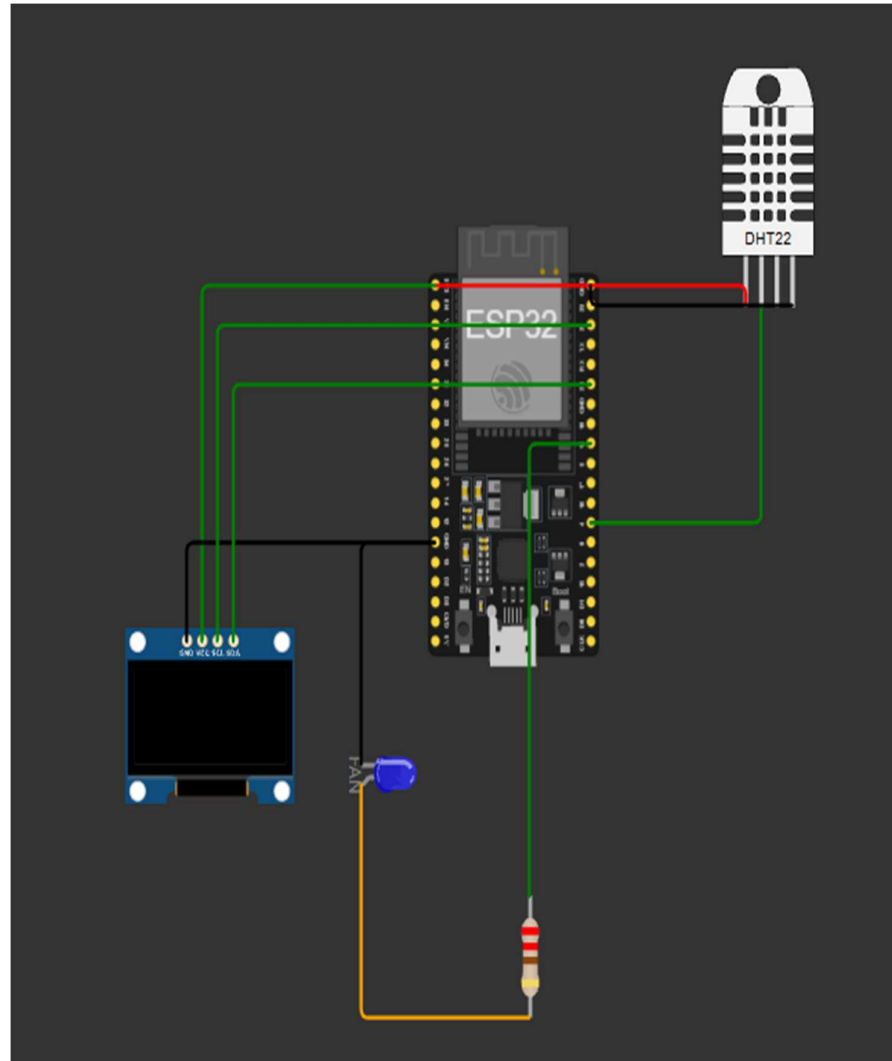
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- simulation and observe the corresponding change in the fan speed.
16. Verify that the fan speed increases when the temperature increases and decreases when the temperature falls.
 17. Stop the simulation after verifying that real-time temperature monitoring and automatic fan-speed control are functioning correctly.

Stepwise Screenshots with steps:





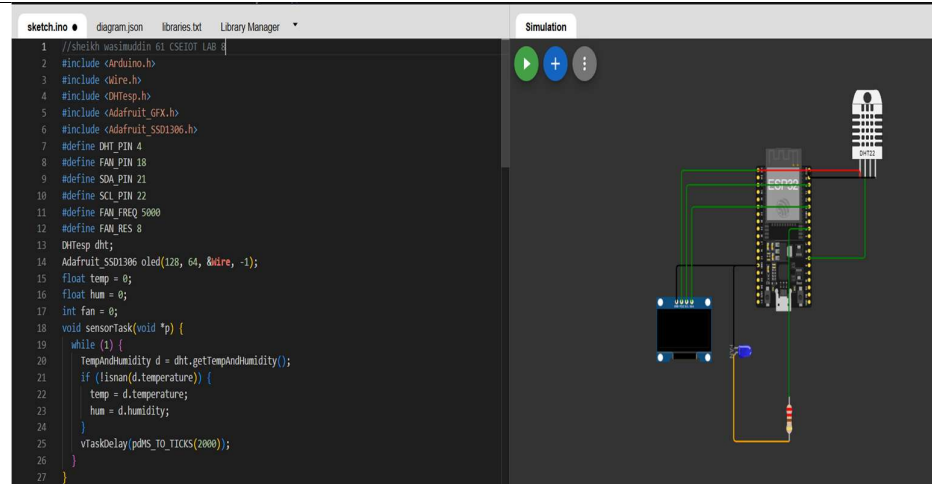
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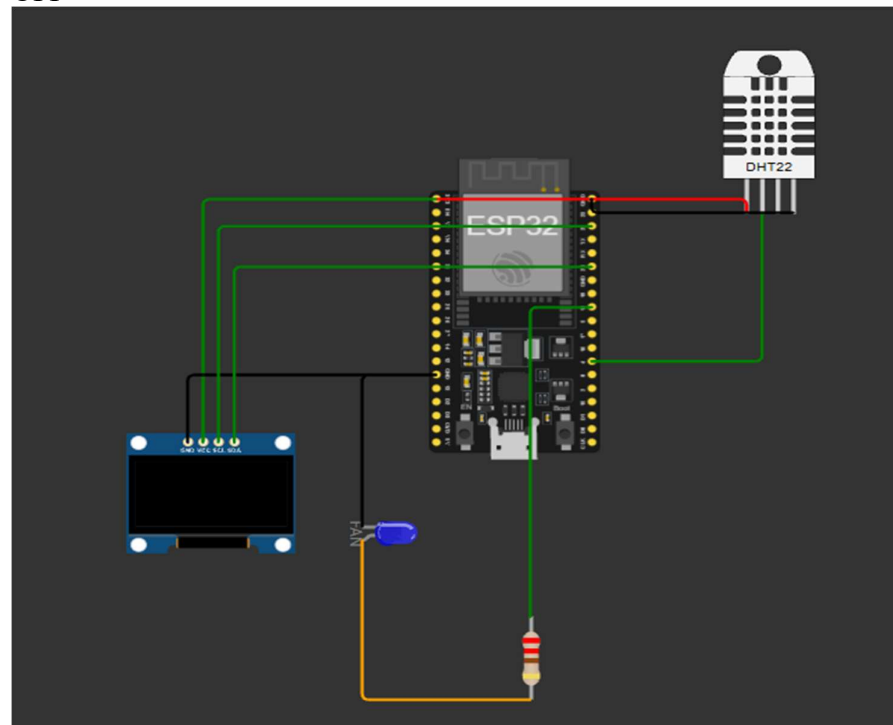
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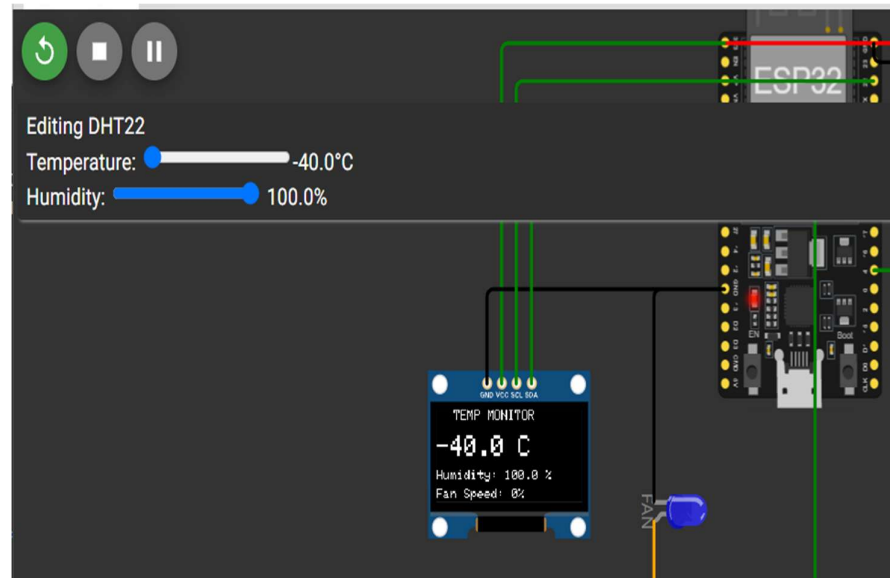
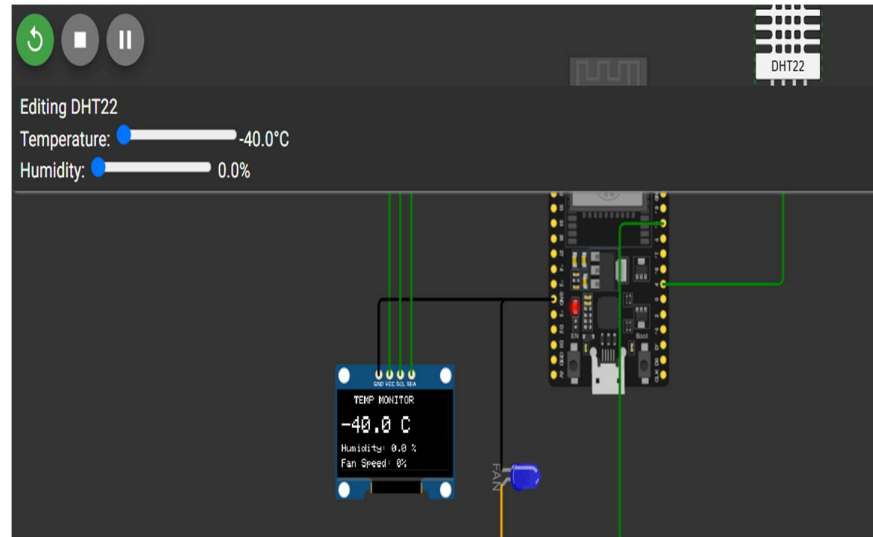
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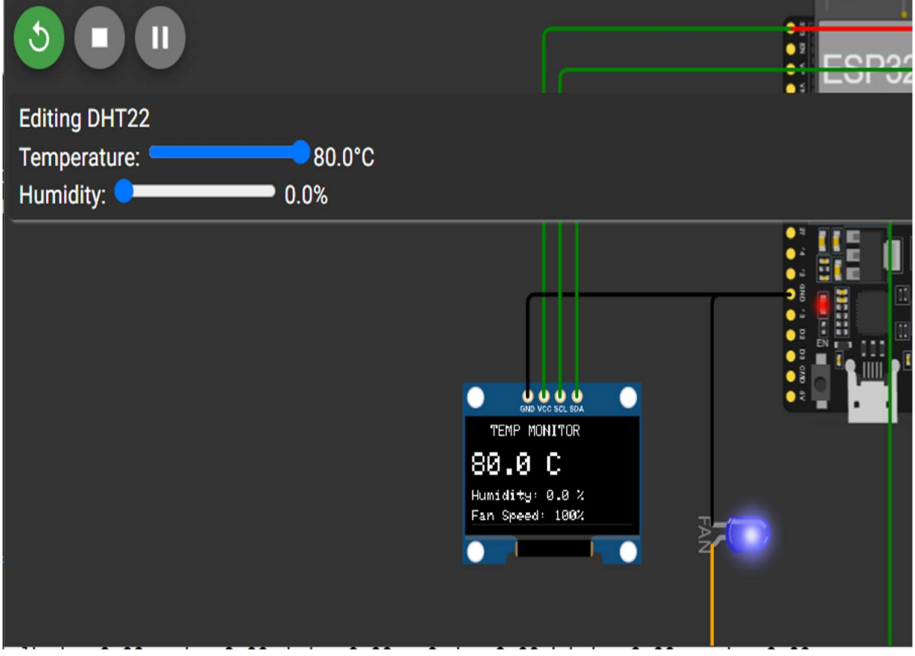
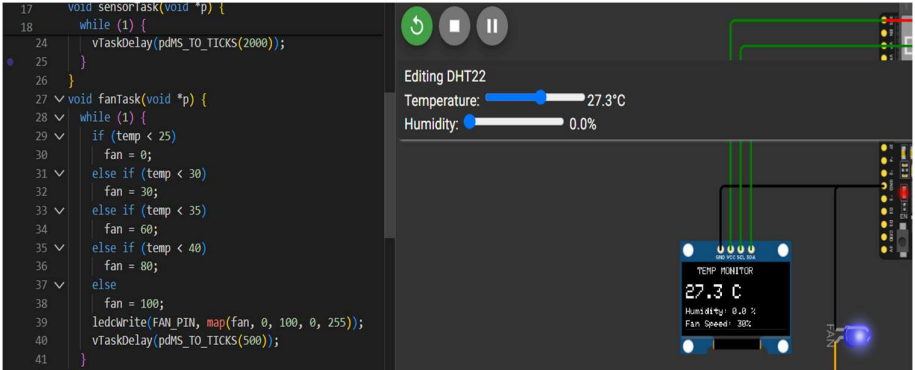
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	  <pre>17 void sensorTask(void *p) { 18 while (1) { 24 vTaskDelay(pdMS_TO_TICKS(2000)); 25 } 26 } 27 void fanTask(void *p) { 28 while (1) { 29 if (temp < 25) 30 fan = 0; 31 else if (temp < 30) 32 fan = 30; 33 else if (temp < 35) 34 fan = 60; 35 else if (temp < 40) 36 fan = 80; 37 else 38 fan = 100; 39 ledwrite(FAN_PIN, map(fan, 0, 100, 0, 255)); 40 vTaskDelay(pdMS_TO_TICKS(500)); 41 } }</pre>
Conclusion	The real-time temperature monitoring and fan speed control system was successfully implemented and simulated using ESP32 and FreeRTOS in Wokwi . The DHT22 sensor was used to continuously monitor the temperature, while FreeRTOS tasks were used to perform temperature monitoring and fan control simultaneously. The fan speed was controlled using PWM according to the measured temperature. The simulation successfully demonstrated automatic adjustment of fan speed with changes in temperature, showing the application of real-time operating systems in embedded systems.
Date	10-09-2026